

Andrei Tregubov

Data Scientist | R&D Engineer

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Date of birth 31 March 2001
Place of birth Nizhny Novgorod



PROFILE

Data Scientist and R&D Engineer with more than 3 years of experience building end-to-end data pipelines and **production-ready ML-models** on large-scale time series data. Expertise with proven impact in signal processing and probabilistic risk modelling applied to real-world data. Author of peer-reviewed publications.

EDUCATION

11/2024 – 03/2026	MSc in Computational Mathematics	University of Passau, Passau, Germany	Grade 1,8
09/2019 – 06/2023	BSc in Fundamental Mathematics	Higher School of Economics, Moscow, Russia	Grade 1,6

PROFESSIONAL EXPERIENCE

07/2021 – 11/2024	Data Scientist & R&D Engineer (Full-time) Institute of Applied Physics Nizhny Novgorod, Russia (Remote) <i>Worked in a small cross-functional research team, driving technical and analytical work.</i> - Independently designed and built an end-to-end sensor data processing application, achieving a 40x reduction in processing time (10h to 15min), replacing fully manual, dataset-specific workflows with a fully automated solution - Engineered a domain-driven data modelling and processing methodology to clean, segment, and extract representative statistical features from over 2.1 billion time-series data points, enabling reliable rare event analysis from long-term field measurements - Developed a machine learning model for predictive risk assessment that correctly predicted 52% of rare hazardous events; subsequently commercialised to improve maritime operational safety - Formulated a novel analytical model for probability estimation of rare events in complex stochastic time-series data - Secured €130,000 in competitive funding and translated complex technical findings into clear presentations for international audiences at 19 conferences and in 2 peer-reviewed publications
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SKILLS

Programming languages	Python, MATLAB, R, SQL
Python libraries	Pandas, NumPy, SciPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch, XGBoost, Statsmodels, SQLAlchemy, Transformers
Tools	Git, Docker, AWS, Linux, Tableau, Jupyter, Flask, REST-APIs, Claude Code
Core expertise	Machine Learning (ML Pipelines), Research and Development, Mathematics, Data Analysis, Predictive Risk Modelling, Data Science, Automation, Statistics, Anomaly Detection, Time-Series Analysis, Signal Processing, Big Data, Hypothesis Testing, Correlation Analysis, Probability Theory, Numerical Methods, Scientific Computing, Data Visualisation, Simulation & Modelling

RESEARCH & PUBLICATIONS

Frequency Spectra and Surface Wave Height Distributions Based on Measurements off Sakhalin Island

Atmospheric and Oceanic Physics, Springer, Oct 2025 | DOI: [10.1134/S0001433825700884](https://doi.org/10.1134/S0001433825700884)
Large-scale analysis of real-world sensor data; statistical modelling of extreme event probability

Probability Distributions for Finite Ensembles of Irregular Waves

Springer Proceedings in Earth and Environmental Sciences, Nov 2023 | DOI: [10.1007/978-3-031-47851-2_19](https://doi.org/10.1007/978-3-031-47851-2_19)
Numerical simulation and probabilistic modelling of anomalous events in time-series data

LANGUAGES	German B2 (Professional working proficiency)	English C2 (Full professional proficiency)	Russian (Native)
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